

Project Design Phase Solution Architecture

Date	22 September 2026
Team ID	SWTID-2026-9721
Project Name	Weather wise
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Example - Solution Architecture Diagram:

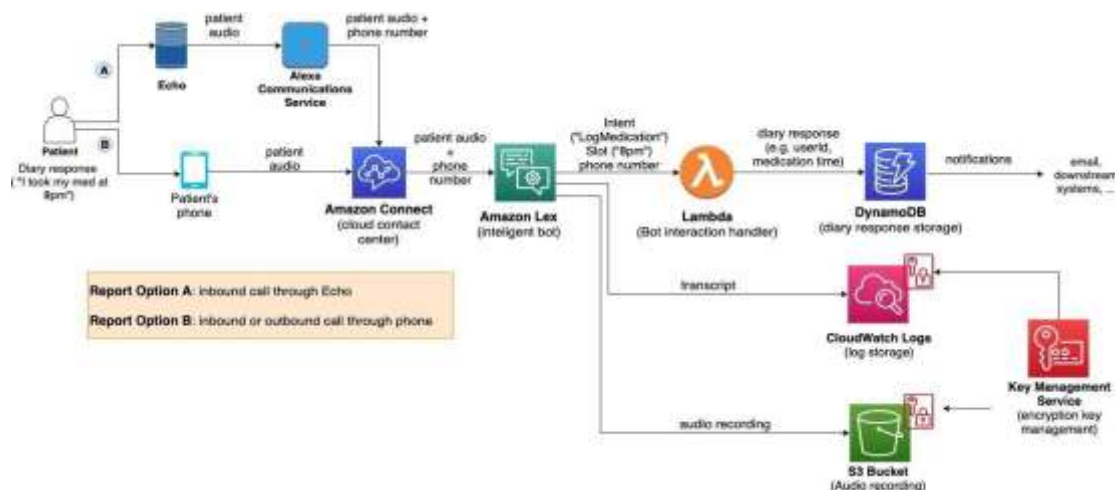


Figure 1: Architecture and data flow of the voice patient diary sample application

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>

Project Design Phase
Proposed Solution Template

Date	22 September 2026
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Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Users may find it difficult to understand changing weather conditions and plan their daily activities accordingly.
2.	Idea / Solution description	Weather Wise provides weather information, weather summaries and personalized recommendations to help users plan their activities.
3.	Novelty / Uniqueness	The system combines weather information with simple and useful recommendations based on the current weather condition.
4.	Social Impact / Customer Satisfaction	It helps users make better daily decisions, stay prepared for changing weather and improves convenience and satisfaction.
5.	Business Model (Revenue Model)	The system can generate revenue through advertisements, premium features and partnerships with weather-related services.

6.	Scalability of the Solution	The solution can be extended to support more locations, more weather features, more users and mobile/web platforms
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Project Design Phase

Problem – Solution Fit Template

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Problem – Solution Fit Template: The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer’s problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ☐ Solve complex problems in a way that fits the state of your customers.
- ☐ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ☐ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ☐ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ☐ **Understand the existing situation in order to improve it for your target group.**

Template:

1. CUSTOMER SEGMENT(S) Who is your customer? (e.g. existing parents of 10-12 y.o. kids)	6. CUSTOMER CONSTRAINTS What constraints prevent your customers from solving a problem or from their desired behavior? (e.g. existing power, budget, etc. needs, personal constraints, available devices)	5. AVAILABLE SOLUTIONS Which solutions are available to the customer when they face the problem or need to get the job done? (e.g. How they used to do the job? What new solutions are available based on paper and pencil or all-digital solution in digital marketing)
2. JOBS-TO-BE-DONE / PROBLEMS What job-to-be-done (or problem) do you address for your customer? (There could be more than one, explore different roles)	9. PROBLEM ROOT CAUSE What is the real reason that this problem exists? What is the back story behind the need to do this job? (e.g. customer's need to do a behavior or the change in regulations)	7. BEHAVIOUR What does your customer do to address the problem and get the job done? (e.g. identify relevant first thought, then great thoughts, opportunities and benefits, identify successful customers, repeat the process, collecting, and so on)
3. TRIGGERS What triggers customer to act? (e.g. seeing their neighbor installing solar panels, reading about a more efficient solution on the news)	10. YOUR SOLUTION Why are you working on an existing business, while your competitors don't? (e.g. in the current, and check how much it fits reality) If you are working on a new business proposition, then how it works and you fit in the market and come up with a solution that has better customer benefits, solves a problem and matches customer behaviour	8. CHANNELS of BEHAVIOUR How do customers do the job? (e.g. online) What kind of actions do customers take online? (e.g. search, click, share, like, etc.)
4. EMOTIONS: BEFORE / AFTER How do customers feel when they face a problem or a job and afterwards? (e.g. how, response, confidence, etc.)	11. OFFLINE What kind of actions do customers take offline? (e.g. search, click, share, like, etc.) How do they feel when they face a problem or a job and afterwards?	

References:

1. <https://www.ideahackers.network/problem-solution-fit-canvas/>
2. <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>